

Optimizing Supply Chain Efficiency in Manufacturing Industries: A

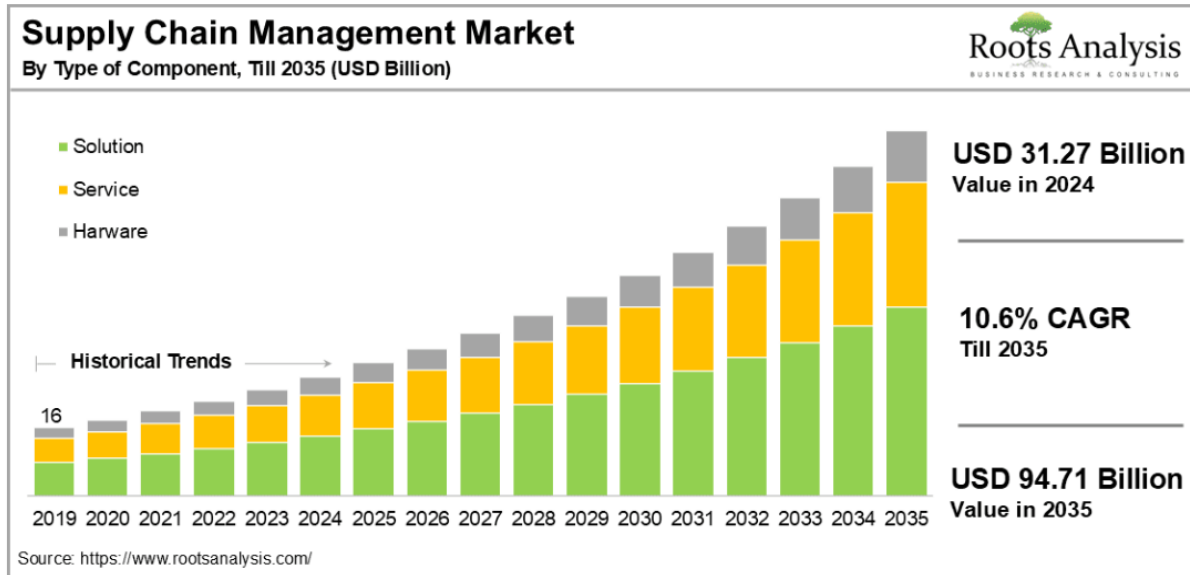
Case Study of Emami

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1. Introduction

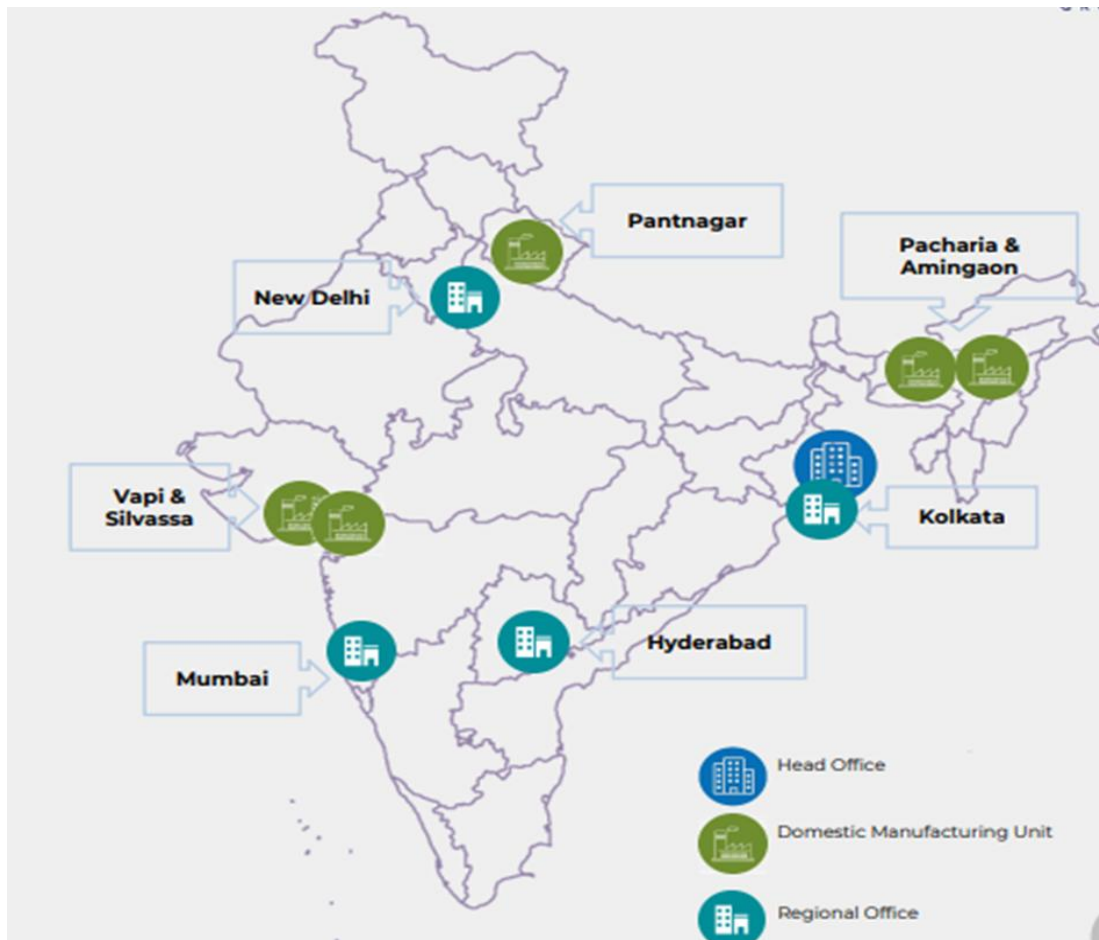
Supply chain management (SCM) has recently become foundational to competitiveness for firms within the manufacturing world. As a result of the globalization of production, manufacturing firms are being pressured to make high-quality products at lower prices faster, with more flexibility, and with more resilience than ever before. With proper supply chains management, firms can move raw materials, intermediate, and finished goods from their suppliers to their customers without delays and with minimal waste. For manufacturing firms, they can no longer focus on just supply chain costs, but also on product availability, market flexibility, and long-term sustainability.

Within the fast moving consumer goods (FMCG) sector, the supply chain has a particularly important role in enabling efficiency, due to the highly competitive nature of FMCG, the breadth of products manufactured, and the variability, and fluctuations in consumer demand. The FMCG sector deals with products where the product life cycle is timed constrained with seasons, demand is seasonal, and the distribution network must also find ways to work across both urban and rural geographies. Each point of impracticality within a firm's supply chain can end up costing both sales and market share, and tarnishing consumer trust. Therefore, optimising the supply chain becomes a strategic imperative to uphold profitability, ensure responsiveness, and build resiliency in the marketplace.



This research examines Emami Limited, an Indian FMCG company that produces a variety of health, personal care, and beauty products, many of which are herbal and Ayurvedic. Emami sells its products in more than 70 countries, it has 5+ domestic manufacturing units, 35+ contract manufacturers located in India, and partners in Bangladesh, United Arab Emirates, Thailand, and Sri Lanka. In fact, Kellogg's ranks Emami among the largest supply networks in the business. Emami has sold its products through over 5.4 million outlets across India with 3,400 distributors and modern trade and e-commerce platforms. Subsequently, Emami can provide a useful case study on how supply chains need to be managed efficiently and effectively in the FMCG sector balance cost, scale, and sustainability.

Manufacturing Footprint



Research questions

- How does Emami perform, manage, procure, and manage logistics in inventory to maximize efficiency?
- What are the risks and bottlenecks in the supply chain?
- What role does sustainability have in the optimization of the supply chain using digital technologies?

2. Literature Review

2.1 Theoretical overview of supply-chain optimization in manufacturing

Supply-chain optimization in manufacturing has traditionally relied on three interrelated paradigms: Lean manufacturing, Just-In-Time (JIT), as well as integrated planning systems like MRP/ERP. Lean manufacturing highlights eliminating waste, continuous improvement, and value for the customer; the core principles and tools of Lean manufacturing are value-stream mapping, pull systems, and continuous improvement (Womack and Jones, 1996). Just-In-Time - made popular in the Toyota Production System - complements Lean manufacturing by emphasizing the right parts, at the right time, so inventory can be minimized and both flow and responsiveness are increased (Ohno, 1988).

Material Requirements Planning (MRP) and its progression through Enterprise Resource Planning (ERP) represents the technology to actually operationalize these two principles of Lean and Just-in-Time (JIT) on scale; while the approaches of Lean and JIT help firms to improve flow and agility, MRP and ERP provide both the backbone, and practice management tools, for executing Lean and JIT principles on scale. MRP and ERP allow firms to move from forecasting demand to ordering materials to executing production schedules with automated planning, and can help mitigate stockouts and overstocking. Moreover, MRP and ERP have gained popularity because of empirically proven outcomes resulting from MRP and ERP implementations in firms. The multiple systematic reviews of the ERP and MRP literature show similar outcomes for the implementations. Among firms implementing ERP, there were improved visibility, smaller inventory on hand, and better coordination of procurement, production, and distribution; that said, improvement results will vary based on implementation quality and change management.

Sustainable supply-chain management emerged as another optimization goal. Academic literature has synthesized conceptual frameworks for integrating environmental and social goals with existing logistical and cost goals. (Seuring and Müller, 2008) reviews the actions firms should enact to adjust supplier review, procurement practices, and product lifecycle decisions that meet regulatory, stakeholders, and sector expectations for sustainability.



Source: [ProcurementTactics](#)

2.2 Challenges in FMCG supply chain: seasonality, volatility, and global sourcing

The supply chain in the fast moving consumer goods (FMCG) sector embodies many complexities over and above the nature of complexities that firms experience in other industrial settings, such as large SKU portfolios; shortening product life cycles; fierce competition among retailers; and uncertain seasonal demand (Research on the Supply Chain of FMCG Industry,

2021). These characteristics amplify the challenges found in supply chain disequilibria like the bullwhip effect, demand forecasting errors and inventory obsolescence. Empirical evidence and simulation studies show that volatility and seasonal demand necessitate responsive planning, flexible capacity (for example, contract manufacturing), and different inventory policies (Schultz et al, 2016).

Many of the other sources of risk firms face in FMCG are similar: fluctuations in the costs of inputs, and to a greater extent for agrarian inputs (for herbal/ayurvedic products) that generate additional risk upstream from agricultural inputs. Research shows that coupling short term tactical capacity flexibility with strategic mechanisms (long term contracts, backward integration) allow firms to reduce costs associated with supply volatility (Wood et al, 2019). Likewise, firms can pursue global sourcing so they benefit constantly from supply (cost advantages, specialty inputs) while exposing their firm to additional risk as well (ex. exchange fluctuations, geopolitical disruption disruptions) which can all increase lead-times.

2.3 Role of technology: ERP, predictive analytics, and AI in supply-chain efficiency

Digital technologies have transitioned from "nice-to-have" to critical tools for supply-chain optimization. ERP/MRP systems are still core to master scheduling, automated procurement, and logistics planning. However, in order for these systems to continue to be effective and efficient, the quality of data, alignment of processes, and governance agreements must be considered (Umble et al. 2003).

In addition to the ERP, predictive analytics and AI (machine learning for demand forecasting, and optimization algorithms for inventory and routing) are adding tangible performance improvement. Indicators suggest that early adopters led to substantial benefits.

Some indications suggest that companies have made improvements across many areas, including inventory levels, reduced logistics costs, and increased service levels while utilizing AI embedded into forecasting and supply-chain decision-making (McKinsey & Company, 2021). Significant gains from using AI to benefit from analytics will also depend on the quality of data inputs, coordination across functions, and pilots that link analytical outputs to operational processes (McKinsey & Company, 2021). All this combined, integrating AI or analytics into their day to day processes, can help companies to plan in a better way.

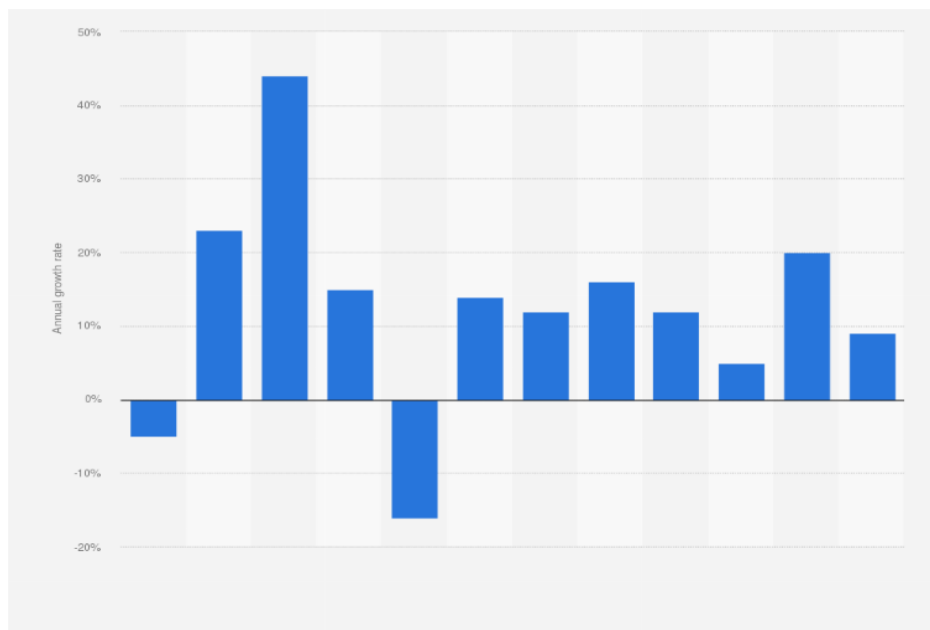
2.4 Comparison with other FMCG players (HUL, Dabur, Patanjali)

Insights from comparisons within large fast-moving consumable goods (FMCG) companies demonstrated real-world learning opportunities. MNC and large national companies (e.g., Hindustan Unilever (HUL) and Unilever) dedicate detailed capital investment to support integrated supply-chain sustainability, local sourcing alternatives to raw materials, and capturing digital tools for distributor networks and equipment used to sense demand — all in published company sustainability reports (Hindustan Unilever Limited, 2024; Unilever, 2024). In this engagement, HUL illustrated examples of an eco-sourcing sustainability approach and a green logistics program that depicted sustainable systems in terms of supply-chain efficiency, and implementation costs trade-offs.

Dabur, as a large manufacturer of Ayurvedic and herbal FMCG products, further illustrates a hybrid sourcing approach to its production by endorsing mix certification of backward integration to some critical herbal ingredients. The hybrid approach further supports Dabur to navigate seasonality challenges, thus creating stability in supply-side uncertainty, and quality. Patanjali's accelerated scale-up is frequently illustrated as an example of aggressive

domestic sourcing and extensive contracting manufacturing and distribution networks — varying levels of progress enhanced by reduced development resources demonstrate both opportunities and risks associated with rapid supply-side network development, brand placement, and sustainability in the herbal FMCG space (Research on the Supply Chain of FMCG Industry 2021 summary).

International performance rate of Emami Limited from financial year 2013 to 2024



Source: Statista

3. Research Methodology

This study employs a case study methodology to explore supply chain optimization in the fast-moving consumer goods (FMCG) sector and in particular of Emami Limited, as the FMCG sector is known to be very complex because of its distribution and logistics management. The

study breathes the use of only secondary sources as its method of collecting data as there were no issues with collection methods, as data in study is secondary data which included the company's internal presentations and reports available to public on the internship through government documentation and literature that referenced these studies published by other academics and professionals in the industry. The triangulation of these sources had also added trustworthiness of findings as it provided a broader composite of thoroughly understanding Emami's supply chain strategy, structure and issues associated

The aimed analysis uses a systematic framework which involved SWOT analysis, risk analysis, and KPI benchmarking, SWOT makes internal strengths and weaknesses visible, and compares them against external threats and opportunities; while risk analysis shows the vulnerabilities, including seasonality related to raw materials and regulation; KPI benchmarking shows performance and efficiency on cost savings on procurement, supplier compliance, and which focuses on days in inventory. Together, the forms of analysis provide a well-rounded consideration of Emami's network of supply chain stakeholders' efficiency.

4. Company & Supply Chain Overview (Emami Case Study)

Emami Limited is one of India's leading fast-moving consumer goods (FMCG) company with a wide array of health and personal care, as well as herbal products in its portfolio. Based in Kolkata, Emami operates with manufacturing sites and distribution points from which it can facilitate its strong domestic and international trade. While its manufacturing base includes five plants in India (its sites also include a manufacturing plant in Bangladesh), Emami purchases products from over 35 contract manufacturers in India and three contract manufacturers located

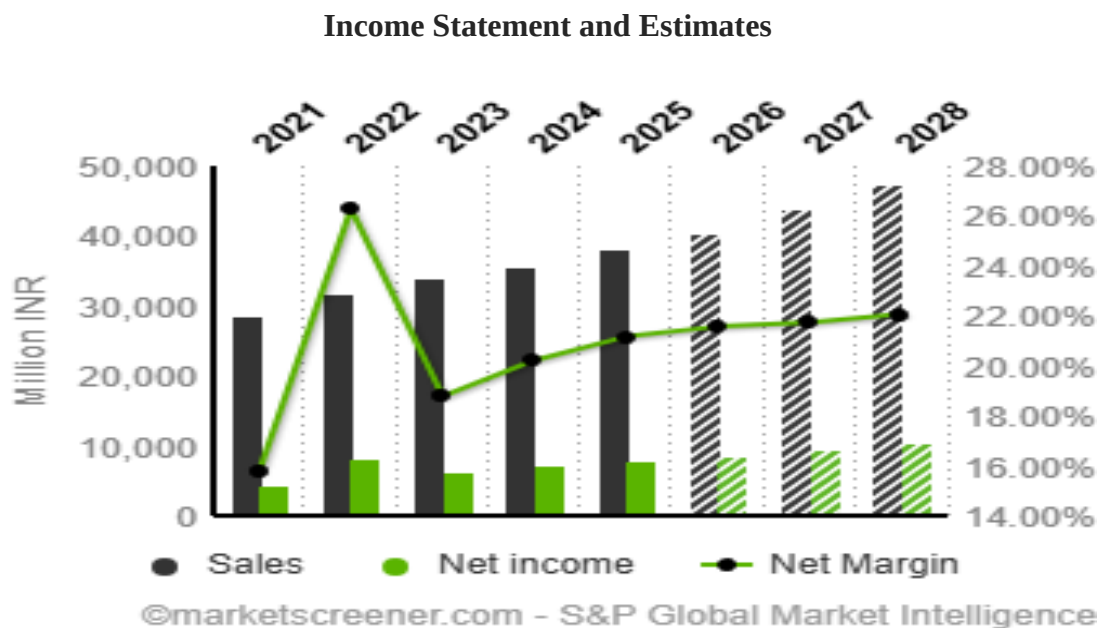
in the UAE, Thailand, and Sri Lanka. This provides the company with the flexibility of production, the ability to respond quickly to sales orders, and the ability to manufacture products that are in close proximity to consumers. Distributing products, Emami and its products are in over 70 countries, and in partnership with approximately 3,400 distributors, it reaches 5.4 million retail outlets in India through modern trade and multiple e-commerce channels. The sophistication and scale of Emami represent the company's high degree of reliance on effective and resilient supply chains.

Metric	Description	Source / Notes
Historical Net Sales (FY'20–FY'25)	Emami's net sales increased from ~₹2,654.88 crore in March 2020 to ~₹3,809.19 crore in March 2025.	Markets Mojo – Shows growth trend; useful baseline for comparing supply chain costs or changes in operational efficiency.
Profit after Tax (Same period)	Profit grew from ~₹308.32 crore (March 2020) to ~₹814.55 crore (March 2025).	Markets Mojo – Indicates possible impact of cost control measures including supply chain optimization.
Q1 FY26 – Profit Growth	9% YoY increase in net profit (₹164.2 crore), despite flat revenue (₹904 crore vs ₹906 crore previous year); attributed to lower input costs & better ops.	The Times of India – Suggests operational / supply chain improvements are having measurable financial impact.

Emami's supply chain management has four parts - procurement, production, warehousing & inventory, and logistics & distribution. For procurement, Emami evaluates the specifications of its raw and packaging materials, deciding whether to source some materials domestically or globally. There is evidence of a dual focus on where to best provide finished goods accessibility for the EoL consumers at both supply points in order to optimize the supply base while maintaining resiliency of supply. Production is done through Emami manufacturing assets and contracted units who produce part of a bulk production order. Each of these allows Emami to scale-up and increase its ability to respond quickly and effectively to spikes and changes in demand.

Emami has a number of key characteristics in their supply chain operations. First, the company is utilizing ERP-based demand forecasting and inventory planning that integrates sales forecasting with production and purchasing plans through an integrated tech connectivity approach. Therefore, Emami's integrated tech approach minimizes the potential understock and overstock associated with inventory while enhancing their ability to be responsive. Second, Emami's procurement and production cadence is seasonal based (M0, M1, M2 cycles), where production plans are filed based on major seasonal upticks in demand (for example, cooling oils in the summer as compared to balms in the winter). This cadence allows for more consistency of work across the sales, operations, and procurement teams. Third, Emami practices dual sourcing and active supplier management to mitigate risks associated with supplier locks and price volatility and to ensure sufficient supply stability, especially in more volatile marketplaces. Finally, Emami sources locally while balancing that volume with 'imports' from some new European and Asian suppliers - in some cases for certain ingredients (e.g., fragrances and

essential oils). Emami utilizes this hybrid approach to support quality, control cost, and satisfy the expectations of their diverse geography of consumers.



Source: [MarketScreener](https://marketscreener.com)

5. Optimization Practices

Emami has taken a number of operationalized supply chain management approaches to optimize supply chain efficiency, manage complexity, mitigate waste and inefficiencies, and enhance the ability to be responsive to changing market conditions in a hyper-competitive FMCG marketplace, all based on five management principles; procurement, inventory, production scheduling, logistics, and sustainability.

Procurement strategies play an integral role in Emami's supply chain optimization. Emami implements a dual sourcing strategy for its key raw and packaging materials. The dual sourcing strategy mitigates reliance on suppliers and further limits risk associated with supply

chain disruptions. Emami continues on building long-term supplier contracts to hedge against cost variability associated with price fluctuation of inputs such as oils, packaging materials, and other key raw materials. Emami works to balance inputs from local and international supply sources in a prudent manner. With local sourcing practices costs savings will accrue and stock replenishment timelines are quicker, and it continues to rely on fragrances and specialty chemicals imported from a global supply base adding another level of assurance quality and differentiation of the product.

In warehouse and inventory management, Emami utilizes EH to monitor inventory status, manage safety stocks, and assist with replenishment. Inventory management is further supported by ABC analysis, allowing visibility into managing inventory of slow-moving products and inventory with the highest priority, ensuring resource allocation is committed as appropriate to high-priority products. Seasonal demand is also of importance, especially in demand peaks where safety stocks are built around peak demand periods, for example, cooling oils in summer and skin-care creams in winter. They also incorporate FIFO and FEFO to reduce waste and obsolescence.

At Emami, the production scheduling process entails a coordinated approach between the ERP system and MRP, which ultimately drives efficient acquisition of resources while maintaining flexibility in production cycles. Monthly S&OP meetings provide an opportunity for production schedules to be set for M0, while also forecasting for M1 and M2. Specifically, production schedules and forecasting help facilitate better sync and coordination between procurement (purchase branches), production units, and contract manufacturers. Therefore, to obtain the fluidity to accommodate unforeseen demand changes, and keep resources aligned on a consistent monthly basis is the optimal management approach.

Emami manages the logistics and distribution of products through the ERP logistics modules for arriving at synchronized movements. Specifically, to synchronize shipments with production output and the needs of the distributor, the finished products must be allocated to the respective Regional Distribution Centers (RDCs) and Carrying & Forwarding Agents (CFA) that apply to national distributions, this phase of supply chain management by Emami well develops deep market penetration strategies through routing last-mile deliveries from distributors to stockists or signed-up retailers through the use of sub-distributors (buying group), and van sales. Therefore, while consolidation at the RDCs from where finished goods are sent to CFAs is maintained to streamline the last-mile operations to several retailers on a national scale, Emami has also significantly relied upon third-party logistics (3PL) partners that facilitate distribution flexibility and scalability in last-mile deliveries.

Sustainability efforts have become an essential part of Emami's supply chain optimization. Emami has continually emphasized eco-friendly and recyclable packaging, and working with its customers who want sustainable products and materials has shortened the time frame for eco-friendly packages. Also, working in partnership with others to farm in a herbal way on important raw materials such as amla, neem, and aloe vera has also proven to be a sustainable practice. Participation in eco-friendly partnerships is also important for some raw materials. Emami encourages local shipment of raw materials to reduce logistics costs and emissions while optimizing supply chain activities with higher levels of environmental and social responsibility.

6. Challenges & Bottlenecks

Nevertheless, Emami's extensive supply chain network is far from optimally efficient, and a variety of challenges/bottlenecks may be inhibiting an efficient and resilient supply chain.

The seasonality of raw materials is a significant challenge, particularly for herbal inputs such as amla, menthol, aloe vera, and neem. All of them depend on agricultural seasons, or climate solely, so they can easily run out of raw material, have extremely inconsistent specifications, or volatility in pricing. Similarly, while price volatility in raw materials is always an immediate concern, the reasons for fluctuating prices are identical to those for base oils, such as coconut oil and mustard oil, pricing for packaging materials such as laminates. These raw materials and base materials are also highly reactive to the price fluctuations of the global commodities cycle; Emami is then exposed to a cost collection pressure that eats away its margins.

If a CMU is a major bottleneck for manufacturing, it heavily contributes to compliance variability; it is possible that capacity may be reduced without any signals, especially during a high season, and there will be all sorts of quality differences depending on the functional location. A CMU does provide a solid source of flexibility and scalability; however, relying on a CMU takes away significant overall control of the manufacturing process.

On the distribution side, even with no structural barriers to the ducting, poor rural market access can lead to systemic barriers. Since a large part of demand in India is based on growing demand from semi-urban and rural areas, supply chain output productivity and effective delivery can be negatively affected by poor infrastructure and transportation logistics costs and delays. These two variables combined will have a role in the delivery effectiveness on supply chain

activities, along with public variables such as sub-distributors and delivery by van. Last mile issues can cause further challenges on supply chain performance due to a combination of complexity and additional investment to overcome logistics challenges

Regulatory risks also pose challenges for Emami. Greater pressure continues to arise from compliance with AYUSH and FSSAI regulations with Herbal and Ayurvedic products. The rules/legislation relating to AYUSH and FSSAI both require increased scrutiny on product approvals, labelling, and quality control; any inability to maintain compliance will ultimately lead to delayed launches and penalties.

Finally, within the FMCG landscape, Emami competes with several large incumbents, such as Hindustan Unilever, Dabur, and Patanjali, who not only pose a challenge to Emami in relation to loss of market share, but also raise the standard in relation to supply chain sustainability, competitiveness, and technology adoption. If Emami is to remain competitive the task remains to continually improve on innovation, cost, and effectiveness in distribution.

7. SWOT Analysis of Emami's Supply Chain

A SWOT analysis provides a structured overview of Emami's supply chain in terms of its internal strengths and weaknesses and its external opportunities and threats. It allows the company to see where strengths are found and what weaknesses might be impacting its efficiency and competitiveness.

Strengths: Emami's supply chain incorporates ERP-based planning systems that help to introduce synchronization between forecasting demand, capacity planning, procurement,

production scheduling, and logistics into one high-level unified system. The digital framework provides a means of coordination across the functions and ultimately contributes to eliminating inefficiencies. Emami has a footprint, specifically domestic manufacturing, international production, and an expansive retail and distribution network that provides scale and flexibility to support multiple markets.

Weaknesses: While there are elements of strength in Emami's supply chain, there are also elements of weakness. One of the most significant weaknesses is that when it comes to sourcing raw and packaging materials primarily consisting of herbal and oil-based materials, there is dependency on supplier concentration where there are a limited number of affected geographies that supply such materials; Dependency on supplier concentration may increase the risk of supply interruptions, and inconsistent quality. There is a significant amount of reliance on the engagement of contract manufacturing units (CMUs) which allows for scaling purposes, though this relies on other businesses which reduces Emami's control and stewardship of quality, compliance, and capacity utilization.

Opportunities: The growing consumer interest in Ayurveda and herbal products in both domestic and global markets is a significant opportunity for Emami to expand its product ranges, and deepen its position in the market. Furthermore, advances in digital transformation technologies (such as predictive analytics or forecasting based on artificial intelligence, and blockchain-enabled traceability) offer the potential to improve supply chain efficiency, resiliency, and consumer trust.

Threats: Emami is clearly faced with threats that are outside of their control including the price volatility of raw materials as associated with global commodity markets, especially oil inputs,

and associated packaging inputs, which are subject to worldwide commodity markets. Another ongoing threat is the competitive intensity from FMCG Leaders, namely Hindustan Unilever, Dabur, and then Patanjali as they all share a significant audiovisual commitment to sustainability efficiency, sustainability broad branding, and sustainability overarching centralization. Even the lack of logistics and supply-chain assurance due to poor rural infrastructure, changes in fuel prices, and time of seasons (i.e., monsoon flooding and delays), threaten logistics in selling and delivery of product.

In summary, Emami has a strong supply chain benefit of having both digital connectivity and a global footprint, but suffers from a key weakness of supplier dependence, and contract manufacturing parties together. Emami will therefore need to stay on top of situations that provide opportunity because of Ayurveda and digital technology, but also continue to monitor and mitigate threats through rapid transitions to a competitive and sustainable future.

8. Benchmarking & Key Performance Indicators (KPIs)

Emami utilizes a comprehensive set of KPIs to assess the performance of its supply chain, which affords measurable contributions to cost efficiencies, dependable suppliers, inventory, and sourcing efficiencies. Emami can then compare its own KPIs against industry trends and best practice benchmarks to close organization-wide gaps to support its competitiveness as an FMCG organization.

Cost and efficiency KPIs illustrate the effectiveness of sourcing and procuring. KPIs such as procurement cost savings and purchase price variance (which is the difference between the

budgeted price and actual price) are important and necessary in monitoring how well Emami responds to fluctuations in input costs, while also solidifying visibility around efficiencies in procurement functions.

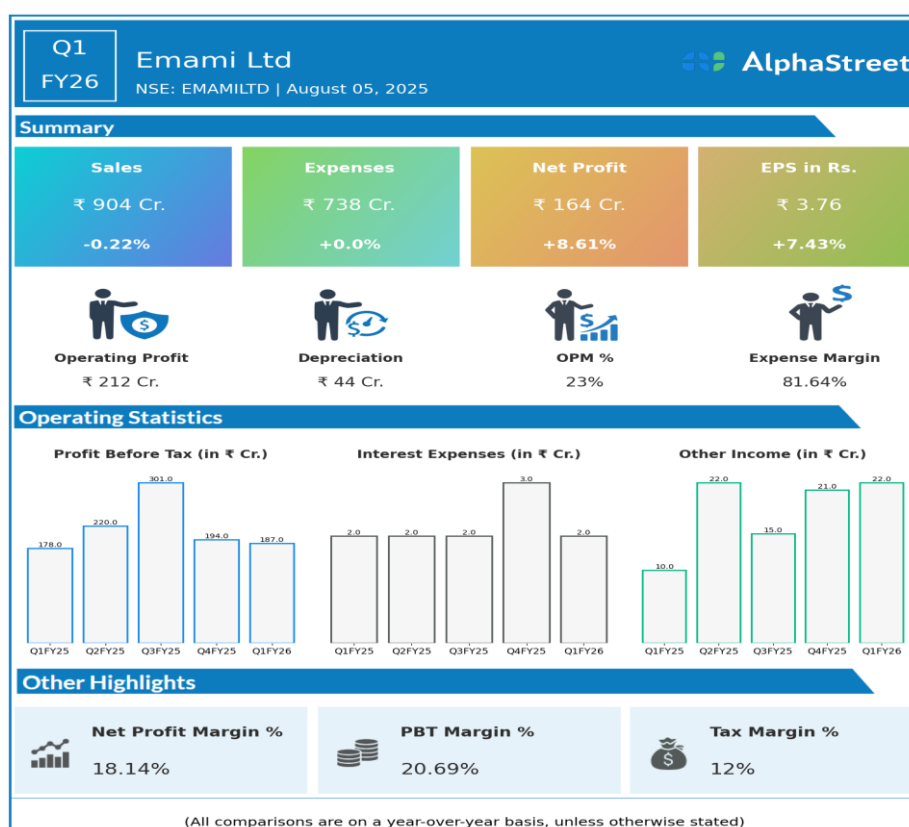
Supplier management KPI's assess the performance of vendor partners. Since Emami also heavily relies on contract manufacturers and 3rd party suppliers, there are multiple KPI components in place to measure the supplier's performance. The important KPI was the supplier OTIF metric that captured the number of times Emami actually received the ordered supplies. The supplier KPI further measures supplier quality compliance standards to ensure that the corresponding quantities of raw materials and packaging conform to exact specifications, to ensure efficiencies that reduce the need to rework, allowing for production time schedules to be maintained.

Inventory KPIs provide information on how responsive stock efficiency and demand planners are to managing inventory. There are KPIs such as inventory holding days that measure the length of time each stock item remains unsold before being purchased by a customer, or if it is a slow-moving stock in the inventory line. There are also fast-moving stock kits that take on many stock items related to the category and take up square footage in the warehouse, mitigating against spoilage of goods and becoming obsolete.

Procurement KPIs record the efficiency and effectiveness of the sourcing and vendor management process. Examples of procurement KPIs will be procurement failure analysis (occurrences of incomplete or delays on orders), the bad stock-rejection percentage (based on what was found when we received all the rejects, which could be raw or packaging material), and procurement cycle time (the time it takes from RFQ to purchase order). Each example then

presents an opportunity for process improvement. In procurement, improving its KPIs will lower transaction costs and create healthy supply partner relationships.

Tracking and benchmarking these KPIs on a regular basis creates more supply chain transparency for Emami. It gives Emami the opportunity to discover and resolve inefficiencies with its own operations and align its own operations to global FMCG supply chain KPIs and strategies. While the resilient supply chain strategy is focused on cost reduction and improving process efficiency, optimization is not just about saving money but also about building a more resilient and responsive supply chain.



Source: [Alphastreet](#)

9. Discussion

Emami's supply chain practices embody the principles of global supply chain efficiency models such as Lean, Just-in-time (JIT), and integrated ERP-driven planning. Emami's dual sourcing, seasonal procurement planning, and ERP-enabled forecasting practices exemplify the best possible practices in manufacturing and FMCG supply chains globally, as it pertains to waste, demand, and lead time responsiveness, and end-to-end integration.

Emami's strategies clearly illustrate convenience while simultaneously establishing trade-offs between costs, availability (Robustness) and sustainability. In the case of Emami, using contract manufacturing illustrates cost efficiency and flexibility, however the degree of control they have around availability (compliance) and assurances of quality is very weak. While sourcing locally offers benefits with lower logistical costs and slightly reduced emissions, the seasonal nature of agriculture (and associated risk responsibilities) adds volatility or stability of availability of raw materials. While modifying packaging to incorporate recycled content demonstrates their commitment to environmental objectives, implementation of available alternative raw plant-based materials may complicate compliance and in some cases, incur additional cost versus competitors. The sustainability partnership with farmers represents an investment by Emami to capture sustainability, although it is likely that there will be little immediate benefit from any potential issues. These trade-offs evidences that in reality, supply chain optimization is not simply about costs, but weighing costs as trade-offs between certain costs versus efficiency, or long-term availability/robustness, and the community responsibility.

The Emami case also offers helpful lessons and insights for other industries in manufacturing. First, companies can reduce dependency risks while meeting quality

specifications with a diversified supply, whether local or global. Second, companies can better align supply with market demand while mitigating inventory risks by employing planned, organized maiden production planning cycles (M0, M1 and M2) that link demand forecasts before production. Third, companies must make property investments into their last-mile and rural area networks in an emerging market (because infrastructure bottlenecks can change their supply chain system). These lessons apply outside of FCMGs and, to that end, are relevant to any sector in manufacturing that is impacted by volatility, seasonality, and distributed demand.

Finally, Emami's experience illustrates the importance of the digital transformation in facilitating efficiencies within supply chains. The company has committed to enterprise resource planning (ERP) and material requirements planning (MRP) to help with synchronized planning among the activities of procurement, production, and logistics functions. However, we know that global best practices are still evolving, and the majority of companies remain deficient with their predictive capacity regarding forecasting, inventory optimization, and logistics planning, especially without predictive analytics and artificial intelligence. Emami can even provide advance indicators to the end user about demand spikes, such as loading their wages, and can help reduce the risks of over-committing from raw material commitments, both helping to provide route optimization for rural logistics. For example, the possibilities associated with blockchain will provide a greater level of confidence regarding era compliance to suppliers for material sourcing, as well as giving consumers a greater level of confidence through traceability for herbal products. Therefore, the digital transformation can be seen as not only improving operational efficiencies, it can also assist in increasing the level of transparency and accountability.

10. Conclusion & Recommendations

This research project investigates supply chain improvement opportunities for Emami in the high-intensity competitive FMCG space to enhance the flexibility, resiliency, and, more importantly, sustainability of professionalism. The company's orientation can be observed in their strategies, all of which include practices of contracting with two suppliers on producing and sourcing product and inventory lines (medium -large volume). They also utilize demand information from their work system (ERP) with which they schedule production, and they maintain their distribution networks well to achieve the best trade-off between cost effective delivery and flexibility. In addition to their industry-leading practices related to cost-effective delivery and planning, Emami has also incorporated a number of environmentally sustainable practices into their supply chain, as sustainability is one of the five orientations of professionalism. Emami's supply chain practices included sourcing for bamboo wrap, accommodating vegetable-based wrap for packaging, and outsourcing herbal farming to use a more authentic, sustainable, and responsible institutional animal production method. As a result of implementing these practices and methods, Emami was able to grow its business, and achieve productivity and flexibility opportunities to enter the domestic and international market.

Based on those findings, a few suggestions might be made to improve Emami's supply chain opportunities associated with sustainability, in particular:

1. A long-term investment in backward integration for critical herbal raw materials will normalise supply input terms. The key benefits associated with backward integration, it will help to reduce reliance on external suppliers for critical inputs and reduce seasonal

volatility in supply consistency for things such as amla, neem, and menthol depending on the contracted production and harvesting schedule.

2. Increase the level of digitalisation through AI-based demand forecasting. Using predictive analytics and artificial intelligence through an ERP can significantly improve the accuracy of demand forecasting, determine appropriate levels of safety stock and mitigate the risk of procurement and distribution.
3. Improve rural logistics through technology-enabled 3PL collaboration. Technology-based digital tools and partnerships in logistics can help with the last-mile dilemma of rural areas which could build better signals, improve reliability, and reduce costs.
4. Explore green logistics for sustainability brand equity. Using electric vehicles through distribution, and using carbon-neutral logistics providers may contribute to Emami's sustainability branding efforts as an FMCG company.

This case is of value to the overall social science supply chain management literature because it serves as an example of a large FMCG organization operating in an emerging market that uses a mix of traditional efficiency-based models, digital systems, and sustainability to create a competitive advantage. This is to say that there are no shortcuts to supply chain advantage, it will not be resolved easily by one tell-tale value (ie cost). Supply chains exist along a continuum of efficiency, resilience, and sustainability.

In summary, recommendation for additional future research includes establishing comparable case studies in FMCG with players such as Hindustan Unilever, Dabur and Patanjali to determine comparative performance practices, and eventually build a combined best practice

framework on impacting FMCG's operational performance. Lastly, conducting comparative studies to other industries may also provide valuable insights of how FMCG supply chain frameworks can emerge from FMCG into other sectors of manufacturing with the same level of challenges pertaining to volatility and seasonality.

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